

Vježba — Izrada mini stranice s React Routerom

U ovoj vježbi polaznici će izraditi malu single-page aplikaciju koja iz korisničke perspektive ima više stranica. Aplikacija neće raditi klasično ponovno učitavanje stranice, nego će koristiti **client-side routing** pomoću `react-router-dom`.

Aplikacija će imati tri glavne stranice:

Home
Courses
About

Uz to, stranica `Courses` će imati linkove prema detaljima pojedinog tečaja, gdje ćemo koristiti dinamički URL parametar.

Cilj vježbe

Cilj je da polaznici uvježbaju:

- instalaciju `react-router-dom` paketa
 - korištenje `BrowserRouter`
 - definiranje ruta pomoću `Routes` i `Route`
 - izradu zajedničkog layouta
 - korištenje `Outlet` komponente
 - navigaciju pomoću `NavLink`
 - označavanje aktivne stranice
 - navigaciju prema dinamičkoj ruti
 - čitanje URL parametra pomoću `useParams`
 - izbjegavanje običnih `<a href>` linkova za unutarnju navigaciju
-

Naziv projekta

08_React/11_Practice_MiniCourseWebsite/

Scenarij

Zamislite da izrađujete malu web stranicu za online akademiju koja nudi nekoliko frontend tečajeva.

Stranica treba imati:

- Home page
- Courses page
- About page
- Course details page

Korisnik treba moći klikati kroz navigaciju bez ponovnog učitavanja stranice.

Na **Courses** stranici treba vidjeti nekoliko tečajeva. Svaki tečaj treba imati gumb ili link **View details**, koji vodi na detaljnu stranicu tog tečaja.

Na detaljnoj stranici treba se iz URL-a pročitati **id** tečaja pomoću **useParams**.

Zadatak

1. Kreirajte novi React projekt

U terminalu pokrenite:

```
npm create vite@latest mini-course-website -- --template react
```

Zatim uđite u projekt:

```
cd mini-course-website
```

Instalirajte pakete:

```
npm install
```

Instalirajte React Router:

```
npm install react-router-dom
```

Pokrenite aplikaciju:

```
npm run dev
```

2. Očistite početni projekt

U `src/App.jsx` obrišite početni sadržaj.

U `src/App.css` obrišite postojeći CSS.

Aplikaciju ćemo sada graditi od nule.

Predložena struktura projekta

U folderu `src` napravite sljedeću strukturu:

```
src/  
  App.jsx  
  App.css  
  main.jsx  
  layout/  
    RootLayout.jsx  
  pages/  
    Home.jsx  
    Courses.jsx  
    CourseDetails.jsx  
    About.jsx  
    NotFound.jsx
```

Ova struktura nije obavezna, ali je dobra navika jer od početka razdvaja layout od pojedinih stranica.

Dio 1 — Priprema podataka

3. Napravite podatke o tečajevima

U ovoj vježbi ne dohvaćamo podatke s API-ja. Koristit ćemo običan array.

Kasnije, kada naučimo više o API-jima i efektima, isti obrazac se može zamijeniti dohvaćanjem podataka sa servera.

U datoteci `src/pages/Courses.jsx` kasnije ćemo koristiti ove podatke:

```
const courses = [
```

```
{
  id: 1,
  title: "React Basics",
  level: "Beginner",
  duration: "4 weeks",
  description:
    "Learn components, JSX, props, state and the basic React mental model."
},
{
  id: 2,
  title: "React Forms",
  level: "Intermediate",
  duration: "3 weeks",
  description:
    "Practice controlled inputs, validation, submit handling and reusable form patterns."
},
{
  id: 3,
  title: "React Router",
  level: "Intermediate",
  duration: "2 weeks",
  description:
    "Build multi-page single-page applications with routes, layouts and URL parameters."
}
];
```

Kasnije ćemo isti array koristiti i u `CourseDetails` komponenti. Za sada ga možete kopirati u obje datoteke, jer još ne uvodimo posebne data module ako to nije obrađeno. Kao dodatni izazov, može se izdvojiti u zasebnu datoteku.

Dio 2 — Izrada stranica

4. Napravite `Home` stranicu

Napravite datoteku:

`src/pages/Home.jsx`

Dodajte:

```
import { Link } from "react-router-dom";
```

```
function Home() {
  return (
```

```

<section className="page">
  <div className="hero">
    <p className="eyebrow">React Learning Path</p>
    <h1>Learn React step by step</h1>
    <p>
      Build a strong foundation in React by learning components, state,
      forms, effects, routing and reusable UI patterns.
    </p>

    <div className="hero-actions">
      <Link className="button primary" to="/courses">
        View Courses
      </Link>

      <Link className="button secondary" to="/about">
        About the Academy
      </Link>
    </div>
  </div>

  <section className="features">
    <article className="feature-card">
      <h2>Beginner friendly</h2>
      <p>
        Lessons start from the basics and gradually introduce more advanced
        topics.
      </p>
    </article>

    <article className="feature-card">
      <h2>Practice based</h2>
      <p>
        Every topic includes exercises so students can immediately apply
        what they learn.
      </p>
    </article>

    <article className="feature-card">
      <h2>Real projects</h2>
      <p>
        The course path focuses on small but realistic applications instead
        of isolated snippets only.
      </p>
    </article>
  </section>
</section>
);
}

```

```
export default Home;
```

Ova stranica je malo bogatija od običnog `h1`. Ima hero sekciju, dva `Link` gumba i tri informativne kartice.

Važno: za internu navigaciju koristimo `Link`, a ne `<a href>`.

5. Napravite `Courses` stranicu

Napravite datoteku:

```
src/pages/Courses.jsx
```

Dodajte:

```
import { Link } from "react-router-dom";
```

```
const courses = [  
  {  
    id: 1,  
    title: "React Basics",  
    level: "Beginner",  
    duration: "4 weeks",  
    description:  
      "Learn components, JSX, props, state and the basic React mental model."  
  },  
  {  
    id: 2,  
    title: "React Forms",  
    level: "Intermediate",  
    duration: "3 weeks",  
    description:  
      "Practice controlled inputs, validation, submit handling and reusable form patterns."  
  },  
  {  
    id: 3,  
    title: "React Router",  
    level: "Intermediate",  
    duration: "2 weeks",  
    description:  
      "Build multi-page single-page applications with routes, layouts and URL parameters."  
  }  
];  
  
function Courses() {
```

```

return (
  <section className="page">
    <div className="page-header">
      <p className="eyebrow">Available courses</p>
      <h1>Choose your next React topic</h1>
      <p>
        Explore a small collection of React courses and open the details page
        for each one.
      </p>
    </div>

    <section className="course-grid">
      {courses.map(course => (
        <article key={course.id} className="course-card">
          <div className="course-meta">
            <span>{course.level}</span>
            <span>{course.duration}</span>
          </div>

          <h2>{course.title}</h2>

          <p>{course.description}</p>

          <Link className="button primary" to={` /courses/${course.id}`}>
            View details
          </Link>
        </article>
      ))}
    </section>
  </section>
);
}

```

export default Courses;

Ovdje se prvi put pojavljuje dinamička navigacija:

to={` /courses/\${course.id}`}>

Ako je `course.id` vrijednost `1`, link vodi na:

`/courses/1`

Ako je `course.id` vrijednost `2`, vodi na:

`/courses/2`

6. Napravite **CourseDetails** stranicu

Napravite datoteku:

src/pages/CourseDetails.jsx

Dodajte:

```
import { Link, useParams } from "react-router-dom";

const courses = [
  {
    id: 1,
    title: "React Basics",
    level: "Beginner",
    duration: "4 weeks",
    description:
      "Learn components, JSX, props, state and the basic React mental model.",
    topics: ["JSX", "Components", "Props", "State"]
  },
  {
    id: 2,
    title: "React Forms",
    level: "Intermediate",
    duration: "3 weeks",
    description:
      "Practice controlled inputs, validation, submit handling and reusable form patterns.",
    topics: ["Controlled inputs", "Validation", "Submit handling", "Reusable forms"]
  },
  {
    id: 3,
    title: "React Router",
    level: "Intermediate",
    duration: "2 weeks",
    description:
      "Build multi-page single-page applications with routes, layouts and URL parameters.",
    topics: ["Routes", "Link", "NavLink", "useParams"]
  }
];

function CourseDetails() {
  const { id } = useParams();

  const courseId = Number(id);

  const course = courses.find(item => item.id === courseId);

  if (!course) {
```



```

return (
  <section className="page">
    <div className="empty-state">
      <h1>Course not found</h1>
      <p>The course you are looking for does not exist.</p>

      <Link className="button primary" to="/courses">
        Back to Courses
      </Link>
    </div>
  </section>
);
}

```

```

return (
  <section className="page">
    <Link className="back-link" to="/courses">
      ← Back to Courses
    </Link>

    <article className="details-card">
      <p className="eyebrow">Course details</p>
      <h1>{course.title}</h1>

      <div className="course-meta details-meta">
        <span>{course.level}</span>
        <span>{course.duration}</span>
      </div>

      <p>{course.description}</p>

      <h2>What you will practice</h2>

      <ul className="topic-list">
        {course.topics.map(topic => (
          <li key={topic}>{topic}</li>
        ))}
      </ul>
    </article>
  </section>
);
}

```

export default CourseDetails;

Ovdje koristimo:

```
const { id } = useParams();
```

`useParams` čita `id` iz URL-a.

Ako je URL:

`/courses/2`

onda je `id` vrijednost:

`"2"`

Važno: `useParams` vraća string. Zato radimo:

```
const courseId = Number(id);
```

Tek tada možemo usporediti s numeričkim `course.id`:

```
item.id === courseId
```

7. Napravite **About** stranicu

Napravite datoteku:

`src/pages/About.jsx`

Dodajte:

```
function About() {  
  return (  
    <section className="page">  
      <div className="page-header">  
        <p className="eyebrow">About us</p>  
        <h1>We teach React through practical examples</h1>  
        <p>  
          Our learning approach focuses on clear explanations, small projects  
          and exercises that build confidence step by step.  
        </p>  
      </div>  
  
      <section className="about-content">  
        <article className="feature-card">  
          <h2>Our approach</h2>  
          <p>  
            We avoid jumping too quickly into advanced topics. Each new concept  
            is introduced only after students have enough foundation to  
            understand it.  
          </p>  
        </article>  
      </section>  
    </section>  
  );  
}
```

```

<article className="feature-card">
  <h2>How students learn</h2>
  <p>
    Students first read and discuss the concept, then build a small
    example, and finally complete a practice task independently.
  </p>
</article>
</section>
</section>
);
}

```

export default About;

Ova stranica nema routing logiku, ali daje polaznicima još jednu rutu i dodatni sadržaj za stiliziranje.

8. Napravite **NotFound** stranicu

Napravite datoteku:

src/pages/NotFound.jsx

Dodajte:

```

import { Link } from "react-router-dom";

function NotFound() {
  return (
    <section className="page">
      <div className="empty-state">
        <h1>Page not found</h1>
        <p>The page you are looking for does not exist.</p>

        <Link className="button primary" to="/">
          Go Home
        </Link>
      </div>
    </section>
  );
}

```

export default NotFound;

Ovu stranicu ćemo koristiti za fallback rutu:

```
<Route path="*" element={<NotFound />} />
```

Dio 3 — Layout i navigacija

9. Napravite zajednički layout

Napravite datoteku:

src/layout/RootLayout.jsx

Dodajte:

```
import { NavLink, Outlet } from "react-router-dom";

function RootLayout() {
  return (
    <div>
      <header className="site-header">
        <div className="brand">React Academy</div>

        <nav className="navigation">
          <NavLink
            to="/"
            className={({ isActive }) =>
              isActive ? "nav-link active" : "nav-link"
            >
            >
            Home
          </NavLink>

          <NavLink
            to="/courses"
            className={({ isActive }) =>
              isActive ? "nav-link active" : "nav-link"
            >
            >
            Courses
          </NavLink>

          <NavLink
            to="/about"
            className={({ isActive }) =>
              isActive ? "nav-link active" : "nav-link"
            >
            >

```

```

        About
      </NavLink>
    </nav>
  </header>

  <main className="main-content">
    <Outlet />
  </main>
</div>
);
}

```

export default RootLayout;

Ovdje koristimo `NavLink` umjesto `Link` zato što želimo aktivnom linku dati posebnu klasu.

Ovaj dio:

```

className={({ isActive }) =>
  isActive ? "nav-link active" : "nav-link"
}

```

znači: ako je link trenutno aktivan, dodaj mu klase `"nav-link active"`. Ako nije, dodaj samo `"nav-link"`.

Također koristimo:

```
<Outlet />
```

`Outlet` označava mjesto gdje će se prikazivati child rute.

Bez `Outlet` komponente, child stranice se neće prikazati.

Dio 4 — Definiranje ruta

10. Definirajte rute u `App.jsx`

U `src/App.jsx` dodajte:

```

import { BrowserRouter, Route, Routes } from "react-router-dom";
import RootLayout from "../layout/RootLayout";
import Home from "../pages/Home";
import Courses from "../pages/Courses";
import CourseDetails from "../pages/CourseDetails";

```

```

import About from "../pages/About";
import NotFound from "../pages/NotFound";
import "../App.css";

function App() {
  return (
    <BrowserRouter>
      <Routes>
        <Route path="/" element={<RootLayout />} />
        <Route index element={<Home />} />
        <Route path="courses" element={<Courses />} />
        <Route path="courses/:id" element={<CourseDetails />} />
        <Route path="about" element={<About />} />
        <Route path="*" element={<NotFound />} />
      </Routes>
    </BrowserRouter>
  );
}

```

export default App;

Ovdje koristimo ugniježdene rute.

Vanjska ruta:

```
<Route path="/" element={<RootLayout />} />
```

prikazuje layout.

Unutar nje definiramo child rute:

```

<Route index element={<Home />} />
<Route path="courses" element={<Courses />} />
<Route path="courses/:id" element={<CourseDetails />} />
<Route path="about" element={<About />} />
<Route path="*" element={<NotFound />} />

```

index ruta znači: kada je korisnik točno na /, prikaži **Home**.

Ruta:

```
path="courses/:id"
```

znači da očekujemo dinamički parametar nakon **/courses/**.

Primjeri:

```
/courses/1
```

Dio 5 — Stiliziranje aplikacije

11. Dodajte CSS

U `src/App.css` dodajte:

```
body {  
  margin: 0;  
  font-family: Arial, sans-serif;  
  background-color: #eef2f7;  
  color: #1f2937;  
}  
  
.site-header {  
  display: flex;  
  align-items: center;  
  justify-content: space-between;  
  max-width: 1040px;  
  margin: 0 auto;  
  padding: 20px 24px;  
}  
  
.brand {  
  font-size: 20px;  
  font-weight: bold;  
}  
  
.navigation {  
  display: flex;  
  gap: 12px;  
}  
  
.nav-link {  
  padding: 8px 12px;  
  color: #344054;  
  text-decoration: none;  
  border-radius: 8px;  
}  
  
.nav-link:hover {  
  background-color: #e9eef6;
```

```
}
```

```
.nav-link.active {  
  background-color: #1f2937;  
  color: white;  
}
```

```
.main-content {  
  max-width: 1040px;  
  margin: 0 auto;  
  padding: 24px;  
}
```

```
.page {  
  animation: fadeIn 0.2s ease-in;  
}
```

```
.hero,  
.page-header,  
.details-card,  
.empty-state {  
  background-color: white;  
  border: 1px solid #d8dee9;  
  border-radius: 16px;  
  padding: 32px;  
}
```

```
.hero {  
  text-align: center;  
}
```

```
.eyebrow {  
  margin: 0 0 8px;  
  color: #475467;  
  text-transform: uppercase;  
  letter-spacing: 0.08em;  
  font-size: 13px;  
  font-weight: bold;  
}
```

```
.hero h1,  
.page-header h1,  
.details-card h1,  
.empty-state h1 {  
  margin-top: 0;  
  font-size: 36px;  
}
```



```
.hero p,  
.page-header p,  
.details-card p,  
.empty-state p {  
  line-height: 1.6;  
}
```

```
.hero-actions {  
  display: flex;  
  justify-content: center;  
  gap: 12px;  
  margin-top: 24px;  
}
```

```
.button {  
  display: inline-block;  
  padding: 10px 16px;  
  border-radius: 8px;  
  text-decoration: none;  
  border: 1px solid #ccd3df;  
}
```

```
.button.primary {  
  background-color: #1f2937;  
  color: white;  
}
```

```
.button.secondary {  
  background-color: white;  
  color: #1f2937;  
}
```

```
.features,  
.about-content,  
.course-grid {  
  display: grid;  
  gap: 20px;  
  margin-top: 24px;  
}
```

```
.features {  
  grid-template-columns: repeat(3, 1fr);  
}
```

```
.about-content {  
  grid-template-columns: repeat(2, 1fr);  
}
```

```
.course-grid {  
  grid-template-columns: repeat(3, 1fr);  
}
```

```
.feature-card,  
.course-card {  
  background-color: white;  
  border: 1px solid #d8dee9;  
  border-radius: 16px;  
  padding: 24px;  
}
```

```
.feature-card h2,  
.course-card h2 {  
  margin-top: 0;  
}
```

```
.course-card p,  
.feature-card p {  
  line-height: 1.6;  
}
```

```
.course-meta {  
  display: flex;  
  gap: 8px;  
  margin-bottom: 16px;  
}
```

```
.course-meta span {  
  padding: 4px 8px;  
  background-color: #f2f4f7;  
  border-radius: 999px;  
  font-size: 13px;  
}
```

```
.details-meta {  
  margin-top: 16px;  
}
```

```
.back-link {  
  display: inline-block;  
  margin-bottom: 16px;  
  color: #1f2937;  
  text-decoration: none;  
}
```

```
.back-link:hover {  
  text-decoration: underline;
```

```
}

.topic-list {
  padding-left: 20px;
}

.topic-list li {
  margin-bottom: 8px;
}

.empty-state {
  text-align: center;
}

@keyframes fadeIn {
  from {
    opacity: 0;
    transform: translateY(4px);
  }

  to {
    opacity: 1;
    transform: translateY(0);
  }
}
```

Očekivano ponašanje aplikacije

Kada korisnik otvori aplikaciju na:

/

treba vidjeti [Home](#) stranicu.

Klik na [Courses](#) treba promijeniti URL u:

/courses

i prikazati listu tečajeva.

Klik na [View details](#) za prvi tečaj treba otvoriti:

/courses/1

i prikazati detalje tečaja [React Basics](#).

Klik na **About** treba otvoriti:

/about

i prikazati About stranicu.

Aktivni link u navigaciji treba biti vizualno označen.

Ako korisnik ručno upiše nepostojeći URL, na primjer:

/something

treba se prikazati **Page not found**.

Puno rješenje

src/App.jsx

```
import { BrowserRouter, Route, Routes } from "react-router-dom";
import RootLayout from "../layout/RootLayout";
import Home from "../pages/Home";
import Courses from "../pages/Courses";
import CourseDetails from "../pages/CourseDetails";
import About from "../pages/About";
import NotFound from "../pages/NotFound";
import "../App.css";
```

```
function App() {
  return (
    <BrowserRouter>
      <Routes>
        <Route path="/" element={<RootLayout />} />
        <Route index element={<Home />} />
        <Route path="courses" element={<Courses />} />
        <Route path="courses/:id" element={<CourseDetails />} />
        <Route path="about" element={<About />} />
        <Route path="*" element={<NotFound />} />
      </Routes>
    </BrowserRouter>
  );
}
```

```
export default App;
```

src/layout/RootLayout.jsx

```
import { NavLink, Outlet } from "react-router-dom";

function RootLayout() {
  return (
    <div>
      <header className="site-header">
        <div className="brand">React Academy</div>

        <nav className="navigation">
          <NavLink
            to="/"
            className={({ isActive }) =>
              isActive ? "nav-link active" : "nav-link"
            }
          >
            Home
          </NavLink>

          <NavLink
            to="/courses"
            className={({ isActive }) =>
              isActive ? "nav-link active" : "nav-link"
            }
          >
            Courses
          </NavLink>

          <NavLink
            to="/about"
            className={({ isActive }) =>
              isActive ? "nav-link active" : "nav-link"
            }
          >
            About
          </NavLink>
        </nav>
      </header>

      <main className="main-content">
        <Outlet />
      </main>
    </div>
  );
}

export default RootLayout;
```

src/pages/Home.jsx

```
import { Link } from "react-router-dom";

function Home() {
  return (
    <section className="page">
      <div className="hero">
        <p className="eyebrow">React Learning Path</p>
        <h1>Learn React step by step</h1>
        <p>
          Build a strong foundation in React by learning components, state,
          forms, effects, routing and reusable UI patterns.
        </p>

        <div className="hero-actions">
          <Link className="button primary" to="/courses">
            View Courses
          </Link>

          <Link className="button secondary" to="/about">
            About the Academy
          </Link>
        </div>
      </div>

      <section className="features">
        <article className="feature-card">
          <h2>Beginner friendly</h2>
          <p>
            Lessons start from the basics and gradually introduce more advanced
            topics.
          </p>
        </article>

        <article className="feature-card">
          <h2>Practice based</h2>
          <p>
            Every topic includes exercises so students can immediately apply
            what they learn.
          </p>
        </article>

        <article className="feature-card">
          <h2>Real projects</h2>
        </article>
      </section>
    </section>
  );
}
```

```

    <p>
      The course path focuses on small but realistic applications instead
      of isolated snippets only.
    </p>
  </article>
</section>
</section>
);
}

export default Home;

```

src/pages/Courses.jsx

```

import { Link } from "react-router-dom";

const courses = [
  {
    id: 1,
    title: "React Basics",
    level: "Beginner",
    duration: "4 weeks",
    description:
      "Learn components, JSX, props, state and the basic React mental model."
  },
  {
    id: 2,
    title: "React Forms",
    level: "Intermediate",
    duration: "3 weeks",
    description:
      "Practice controlled inputs, validation, submit handling and reusable form patterns."
  },
  {
    id: 3,
    title: "React Router",
    level: "Intermediate",
    duration: "2 weeks",
    description:
      "Build multi-page single-page applications with routes, layouts and URL parameters."
  }
];

function Courses() {
  return (
    <section className="page">

```

```

<div className="page-header">
  <p className="eyebrow">Available courses</p>
  <h1>Choose your next React topic</h1>
  <p>
    Explore a small collection of React courses and open the details page
    for each one.
  </p>
</div>

<section className="course-grid">
  {courses.map(course => (
    <article key={course.id} className="course-card">
      <div className="course-meta">
        <span>{course.level}</span>
        <span>{course.duration}</span>
      </div>

      <h2>{course.title}</h2>

      <p>{course.description}</p>

      <Link className="button primary" to={` /courses/${course.id}`}>
        View details
      </Link>
    </article>
  ))}
</section>
</section>
);
}

export default Courses;

```

src/pages/CourseDetails.jsx

```

import { Link, useParams } from "react-router-dom";

const courses = [
  {
    id: 1,
    title: "React Basics",
    level: "Beginner",
    duration: "4 weeks",
    description:
      "Learn components, JSX, props, state and the basic React mental model.",
    topics: ["JSX", "Components", "Props", "State"]
  }
]

```



```

},
{
  id: 2,
  title: "React Forms",
  level: "Intermediate",
  duration: "3 weeks",
  description:
    "Practice controlled inputs, validation, submit handling and reusable form patterns.",
  topics: ["Controlled inputs", "Validation", "Submit handling", "Reusable forms"]
},
{
  id: 3,
  title: "React Router",
  level: "Intermediate",
  duration: "2 weeks",
  description:
    "Build multi-page single-page applications with routes, layouts and URL parameters.",
  topics: ["Routes", "Link", "NavLink", "useParams"]
}
];

```

```

function CourseDetails() {
  const { id } = useParams();

  const courseId = Number(id);

  const course = courses.find(item => item.id === courseId);

  if (!course) {
    return (
      <section className="page">
        <div className="empty-state">
          <h1>Course not found</h1>
          <p>The course you are looking for does not exist.</p>

          <Link className="button primary" to="/courses">
            Back to Courses
          </Link>
        </div>
      </section>
    );
  }

  return (
    <section className="page">
      <Link className="back-link" to="/courses">
        ← Back to Courses
      </Link>
    </section>
  );
}

```

```

<article className="details-card">
  <p className="eyebrow">Course details</p>
  <h1>{course.title}</h1>

  <div className="course-meta details-meta">
    <span>{course.level}</span>
    <span>{course.duration}</span>
  </div>

  <p>{course.description}</p>

  <h2>What you will practice</h2>

  <ul className="topic-list">
    {course.topics.map(topic => (
      <li key={topic}>{topic}</li>
    ))}
  </ul>
</article>
</section>
);
}

```

```
export default CourseDetails;
```

src/pages/About.jsx

```

function About() {
  return (
    <section className="page">
      <div className="page-header">
        <p className="eyebrow">About us</p>
        <h1>We teach React through practical examples</h1>
        <p>
          Our learning approach focuses on clear explanations, small projects
          and exercises that build confidence step by step.
        </p>
      </div>

      <section className="about-content">
        <article className="feature-card">
          <h2>Our approach</h2>
          <p>
            We avoid jumping too quickly into advanced topics. Each new concept
            is introduced only after students have enough foundation to

```

```

        understand it.
      </p>
    </article>

    <article className="feature-card">
      <h2>How students learn</h2>
      <p>
        Students first read and discuss the concept, then build a small
        example, and finally complete a practice task independently.
      </p>
    </article>
  </section>
</section>
);
}

export default About;

```

src/pages/NotFound.jsx

```

import { Link } from "react-router-dom";

function NotFound() {
  return (
    <section className="page">
      <div className="empty-state">
        <h1>Page not found</h1>
        <p>The page you are looking for does not exist.</p>

        <Link className="button primary" to="/">
          Go Home
        </Link>
      </div>
    </section>
  );
}

export default NotFound;

```

src/App.css

```

body {
  margin: 0;
  font-family: Arial, sans-serif;
  background-color: #eef2f7;
}

```

```
color: #1f2937;  
}
```

```
.site-header {  
display: flex;  
align-items: center;  
justify-content: space-between;  
max-width: 1040px;  
margin: 0 auto;  
padding: 20px 24px;  
}
```

```
.brand {  
font-size: 20px;  
font-weight: bold;  
}
```

```
.navigation {  
display: flex;  
gap: 12px;  
}
```

```
.nav-link {  
padding: 8px 12px;  
color: #344054;  
text-decoration: none;  
border-radius: 8px;  
}
```

```
.nav-link:hover {  
background-color: #e9eef6;  
}
```

```
.nav-link.active {  
background-color: #1f2937;  
color: white;  
}
```

```
.main-content {  
max-width: 1040px;  
margin: 0 auto;  
padding: 24px;  
}
```

```
.page {  
animation: fadeIn 0.2s ease-in;  
}
```

```
.hero,  
.page-header,  
.details-card,  
.empty-state {  
  background-color: white;  
  border: 1px solid #d8dee9;  
  border-radius: 16px;  
  padding: 32px;  
}
```

```
.hero {  
  text-align: center;  
}
```

```
.eyebrow {  
  margin: 0 0 8px;  
  color: #475467;  
  text-transform: uppercase;  
  letter-spacing: 0.08em;  
  font-size: 13px;  
  font-weight: bold;  
}
```

```
.hero h1,  
.page-header h1,  
.details-card h1,  
.empty-state h1 {  
  margin-top: 0;  
  font-size: 36px;  
}
```

```
.hero p,  
.page-header p,  
.details-card p,  
.empty-state p {  
  line-height: 1.6;  
}
```

```
.hero-actions {  
  display: flex;  
  justify-content: center;  
  gap: 12px;  
  margin-top: 24px;  
}
```

```
.button {  
  display: inline-block;  
  padding: 10px 16px;
```

```
border-radius: 8px;
text-decoration: none;
border: 1px solid #ccd3df;
}
```

```
.button.primary {
background-color: #1f2937;
color: white;
}
```

```
.button.secondary {
background-color: white;
color: #1f2937;
}
```

```
.features,
.about-content,
.course-grid {
display: grid;
gap: 20px;
margin-top: 24px;
}
```

```
.features {
grid-template-columns: repeat(3, 1fr);
}
```

```
.about-content {
grid-template-columns: repeat(2, 1fr);
}
```

```
.course-grid {
grid-template-columns: repeat(3, 1fr);
}
```

```
.feature-card,
.course-card {
background-color: white;
border: 1px solid #d8dee9;
border-radius: 16px;
padding: 24px;
}
```

```
.feature-card h2,
.course-card h2 {
margin-top: 0;
}
```

```
.course-card p,  
.feature-card p {  
  line-height: 1.6;  
}
```

```
.course-meta {  
  display: flex;  
  gap: 8px;  
  margin-bottom: 16px;  
}
```

```
.course-meta span {  
  padding: 4px 8px;  
  background-color: #f2f4f7;  
  border-radius: 999px;  
  font-size: 13px;  
}
```

```
.details-meta {  
  margin-top: 16px;  
}
```

```
.back-link {  
  display: inline-block;  
  margin-bottom: 16px;  
  color: #1f2937;  
  text-decoration: none;  
}
```

```
.back-link:hover {  
  text-decoration: underline;  
}
```

```
.topic-list {  
  padding-left: 20px;  
}
```

```
.topic-list li {  
  margin-bottom: 8px;  
}
```

```
.empty-state {  
  text-align: center;  
}
```

```
@keyframes fadeIn {  
  from {  
    opacity: 0;  
  }  
}
```

```
transform: translateY(4px);
}

to {
  opacity: 1;
  transform: translateY(0);
}
}
```

Dodatni izazovi

Izazov 1: Dodajte četvrti tečaj

Dodajte novi tečaj u `courses` array i provjerite pojavljuje li se na `Courses` stranici.

Zatim otvorite njegov detail URL, na primjer:

`/courses/4`

Izazov 2: Dodajte `Contact` stranicu

Dodajte četvrtu glavnu stranicu:

Contact

Potrebno je:

1. napraviti `src/pages/Contact.jsx`
2. dodati `NavLink` u `RootLayout`
3. dodati rutu u `App.jsx`

Ovo je odlična provjera razumiju li polaznici cijeli routing flow.

Izazov 3: Uredite aktivni link za detail stranicu

Kada je korisnik na:

`/courses/1`

link `Courses` i dalje treba izgledati aktivno.

Ako se to ne dogodi automatski, možete koristiti [NavLink](#) opcije ili provjeriti ponašanje trenutne verzije React Routera.

Izazov 4: Izdvojite podatke u zasebnu datoteku

Trenutno se `courses` array ponavlja u `Courses.jsx` i `CourseDetails.jsx`.

Napravite datoteku:

`src/data/courses.js`

U nju stavite:

```
export const courses = [  
  // course objects  
];
```

Zatim u obje komponente importajte:

```
import { courses } from "../data/courses";
```

Ovo smanjuje dupliciranje i uvodi bolju organizaciju projekta